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Tut - 3

1) All permutation of array A.

→ Permutation are n! fix one at position '0' & find all recursively n-1! then ~~Stop~~ swap.

F^n becomes

$$P(n) = n \cdot P(n-1) \; ; \; P(1) = 1$$

$$\Downarrow$$

$$P(n) = n!$$

P → array
n → elements

```
Permutate (P, start, n)
    if start = n-1
        output (P);
        return.

    for i = start, n-1
        swap (P[start], P[i])
        Permutate (P, start+1, n)
```

Pyar ka Tohfa, Suraksha ka Vaada

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swap (P[start], P[i])

Time complexity

At depth 'd', there are $\frac{n!}{(n-d)!}$ nodes

$$T(n) = \sum_{d=0}^{n-1} \frac{n!}{(n-d)!} \cdot O(1) + n! \, [O(n)]$$

$$\boxed{T(n) = O(n.n!)}$$

Ex.

$6 \rightarrow 1 + 2 + 3$

$6 \rightarrow$ divisors $1, 2, 3$

$28 \rightarrow 1 + 2 + 4 + 7 + 14$

To get lesser complexity we can use

Euler's Euclid + Lucas Lehmer

(Only for even no.)

For Odd no. [check in future work]

* Reminder.

Algorithm

IsPerfect (n);

if n <= 1;

return False;

divisor_sum = 1;

t = 2

while t * t <= n;

n % t = 0;

divisor_sum = divisor_sum +

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if $t != n/t$;

divisor_sum = divisor_sum + (n/t);

$t = t + 1$

return (divisor_sum) == n.

Loop runs $t-2$ to t

$= \sqrt{n}$.

$\boxed{\text{Time complexity} = \sqrt{n}}$

) Powerset of S

→ if $S = \{a, b, c\}$

$P(S) = \{$ null, $\{a\}, \{b\}, \{c\}, \{a, b\},$
$\{a, c\}, \{b, c\}, \{a, b, c\}\}$

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powerset (S, idex, curr_Subset, result);

if index = |S|;

Add curr_Subset to result
return

powerset (S, index+1), curr_subset, result)

curr_subset = [curr_subset] ∪ {S [index]}

Powerset (S, index + 1, curr_Subset, result)

curr_subset = curr_subset \
S{index}

Time complexity = $O(n.2^n)$

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4) Ackerman's

$$A(m,n) = \begin{cases} n+1 & m=0 \\ A(m-1, 1) & n=0 \\ A(m-1, A(m, n-1)) & \text{or else} \end{cases}$$

Non recursive

```
Ack (m, n)
  stack = (empty);
  push (stack, m)
  curr_n = n
  while stack ≠ empty
      curr_m = pop (stack)
      if curr_m = 0
          curr_n = n+1
      elif
          curr_n = 0
          push (stack, curr_m)
          curr_n = 1
```

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```
else
push (stack, curr_m - 1)
push (stack, curr_m)
curr_n = curr_n - 1
return curr_n.
```

Recursive

```
Ack (m, n)
    if m == 0
        return n+1
    elif n == 0
        return Ack (m-1, 1)
    else
        return (Ack [m-1,
                Ack(m, n-1)]
```

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) Fibonacci Numbers

Recursive

```
Fib (i)
if i==0          return 0;
if i==1          return 1;
else
return Fib(i-1)+ Fib(i+2).
```

Iterative

```
Fib (i)
if i==0          return 0;
if i==1          return 1;
prev 2 = 0
prev 1 = 1
for k = 2 to i
    curr = prev 1 + prev 2
    prev 2 = prev 1
    prev 1 = curr
return curr.
```

LIC ke saath Zindagi rahe Aasaan